3 Computational thinking: Making a good plan

You will learn

- how to plan a program
- what an algorithm is
- what happens when you run a program.

In this unit you will learn to make and use a plan. A plan sets out the steps you have to take to solve a problem. Good planning can help you to achieve your goals.



Class activity

Another name for the plan for a task is an algorithm.

- What is your favourite food? Do you know how to make that food? Write down the actions in the right order.
- What is your favourite computer game? Write down what happens in the game. Put the actions in the right order.

These are examples of algorithms. Make an algorithm into a poster for the classroom wall.



Learning outcomes: Say what an algorithm is and what running a program means

Talk about...

Share the recipe plans and game plans you made. Work together to make the plans even better. Can you add more actions? Can you add pictures?

algorithm
depends on
run a program
command



Did you know?

On internet sites, such as YouTube, you can watch videos of people cooking their favourite foods.



3.1 What actions to take

In this lesson

You will learn:

- how to make a plan by choosing necessary actions.

Spiral back

Last year you used a program made of blocks. Each block stood for one action. Now you will learn to plan a program by putting actions in the right order.

The story of the donkey's hat

There was once an old lady who felt a little sad.



She made a hat for her donkey.

The old lady no longer felt sad.

Her grandson saw the donkey's hat. He asked her,
"How did you do it?"



Activity



Here are some actions that the old lady did that day:

- Peel the potatoes
- Sew flowers on the hat
- Find a straw hat
- Pick flowers
- Drink a glass of water
- Tidy the room
- Put the hat on the donkey's ears
- Make holes in the hat

Which actions are **necessary** to make the hat?
Necessary means actions that must happen.
'Tidy the room' is not necessary to make a hat.

Think again

Write down the things the old lady did. Only write the ones that are necessary to make the hat.

Extra challenge

Write the story of the donkey's hat in your own words. Draw a picture to go with the story.

3.2 The right order

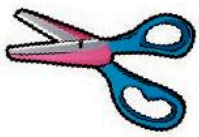
In this lesson

You will learn:

- how to make a plan by putting actions in the right order.

In the last lesson you picked necessary actions to make the donkey's hat. Here are some necessary actions.

Make holes in the hat



Sew flowers on the hat



Put the hat on the donkey's ears



Pick flowers



Find a hat



Now you will put the actions in the right order.

The donkey's hat fits on the donkey's ears. But first you have to make holes, so the ears can fit.

Putting the hat on the donkey **depends** on holes for ears.

One action uses the results of the other action. That's what 'depends on' means.



So, we can tell that

Make holes in the hat

must come before

Put the hat on the donkey's ears

The right order is called the **sequence** of actions.

Activity

Here are two actions:



Sew flowers on the hat



Pick some flowers

Which one has to come first? Say how you know.

Think again

Read all the actions on this page. Write the actions in the right sequence.

Extra challenge

Sometimes two actions are **independent**. That means they do not affect each other. It does not matter which comes first. What two actions in this story are independent? Say how you know.

3.3 An algorithm is a plan

In this lesson

You will learn:

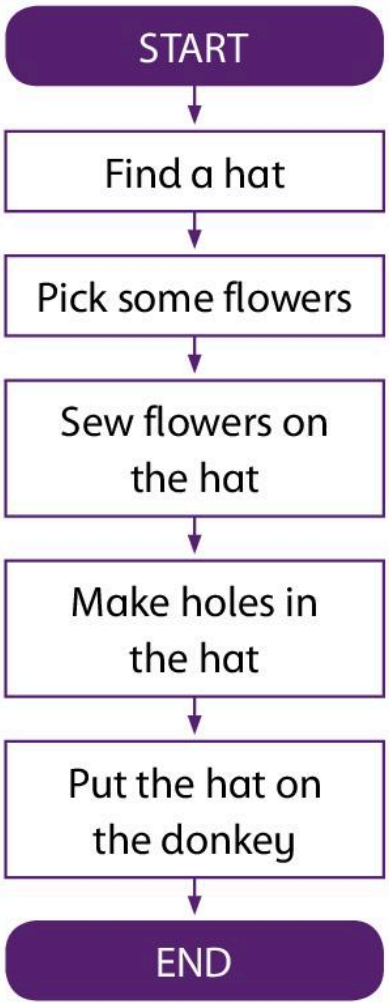
➔ what an algorithm is.

A plan that tells you how to do a task is called an **algorithm**.

The algorithm must include all the actions. The actions must be in the right order.

You can draw an algorithm. Put the actions in boxes. Join the boxes with arrows to show the right order.

Here is an algorithm for making a donkey's hat.



Activity



A big brother made a toy horse for his little sister. Here are the actions.

Find some wood



Carve the wood into a horse shape



Let the paint dry

Paint the horse



Buy some paint

Write the actions in the right order.

Extra challenge

Make an algorithm for making a toy horse. Write the actions in boxes joined with arrows. On a big piece of paper make a colourful picture.

Think again

What does 'algorithm' mean? Write or say the answer in your own words.

3.4 Algorithms and programs

In this lesson

You will learn:

- how making an algorithm helps you to plan a program.



Program

A **program** is a series of commands. The commands tell the computer what to do. When you **run a program** the computer carries out all the commands in the program.

A **programmer** is the person who makes a computer program.

A programmer sometimes makes an algorithm before they make the program. The algorithm is the plan for the program.

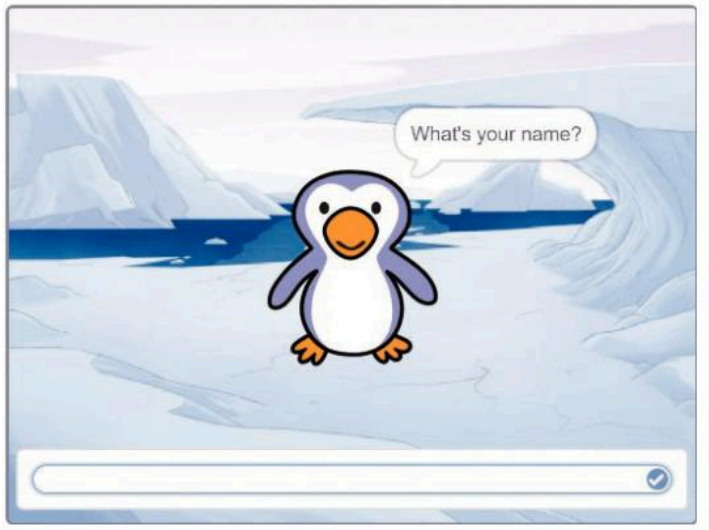
Sprite

You can make a program. You use a programming language to make programs. Scratch is a programming language.

- A Scratch program controls a **sprite**. A sprite is an object or creature on the screen. The program makes it do different things.
- A Scratch program is also called a **script**. A script is a short program.

Activity

A programmer wanted to make a program. The sprite is a penguin. Here is a list of things that the penguin will do.



Say goodbye

Go off the screen

Say hello and your name

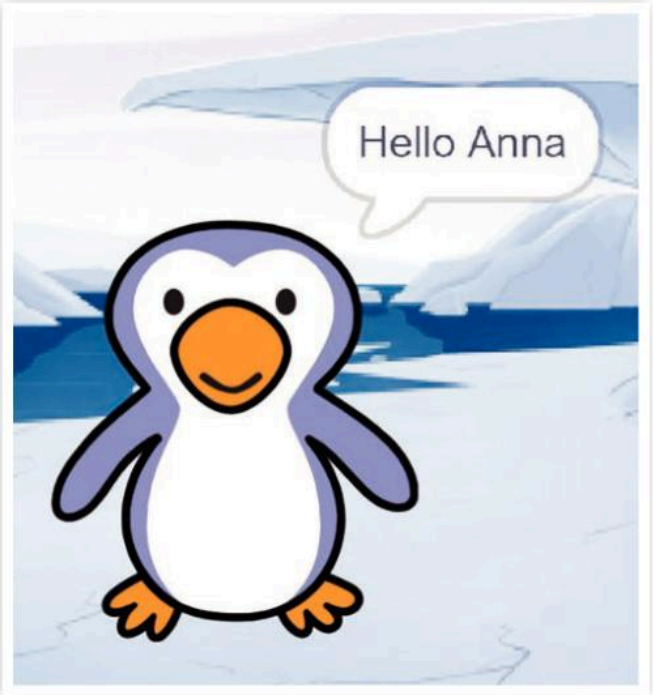
Ask your name

Come onto the screen

Think again

Write an algorithm showing the actions of the program in boxes joined by arrows.

Put these actions in the right order.



Extra challenge

Think of one extra action the penguin could do. Make a plan for the program that includes the extra action.

Be creative

Plan a new computer program:

- Draw the sprite.
- Say what actions the sprite will do.

3.5 Plan the Frog Hop Game

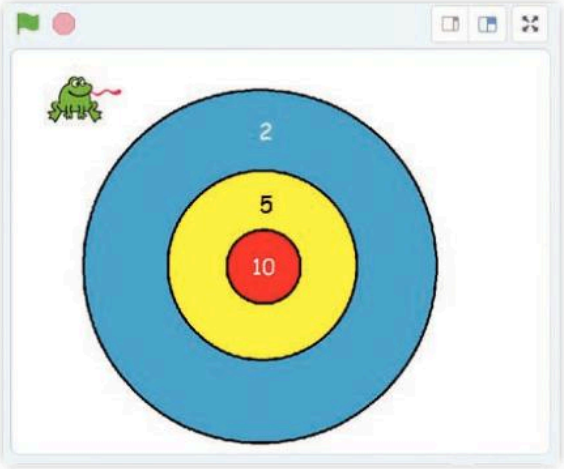
In this lesson

You will learn:

- how to plan a program called the Frog Hop Game.

Program actions

A student decided to make a computer game.



First he wrote down what the game would do.

The sprite is a frog. The background is a target. When you click on the frog it jumps to a new place on the target. You get points depending on where the frog jumps.

Here are the actions of the program:

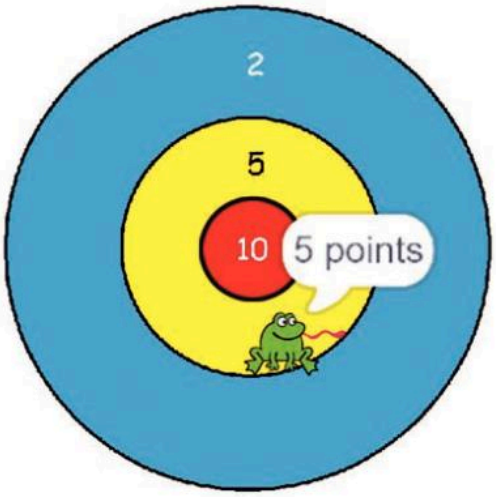
The frog jumps to a place on the target

The user clicks the frog

The frog says how many points you got

Work out how many points you get

Activity



Think again

Here are two actions from the Frog Hop Game.

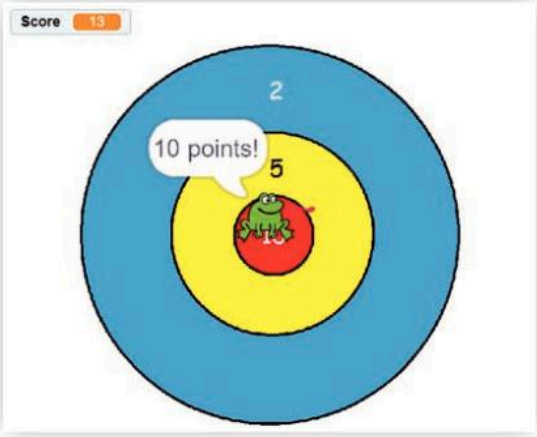
Say how many points	Work out how many points

One action must come before the other. Which one comes first? Give a reason for your answer.

Put the actions in the right order.

Make an algorithm.

Extra challenge



The student wanted to change the program like this.

- Start with total score 0.
- When you get points add to your total score.
- See the total score at the end of the game.

Make an algorithm that includes these new actions.

3.6 Run programs

In this lesson

You will learn:

- how to run a program.

Load the program

There is a ready-made program for you to **run**. 'Run a program' means the computer carries out the commands in the program.

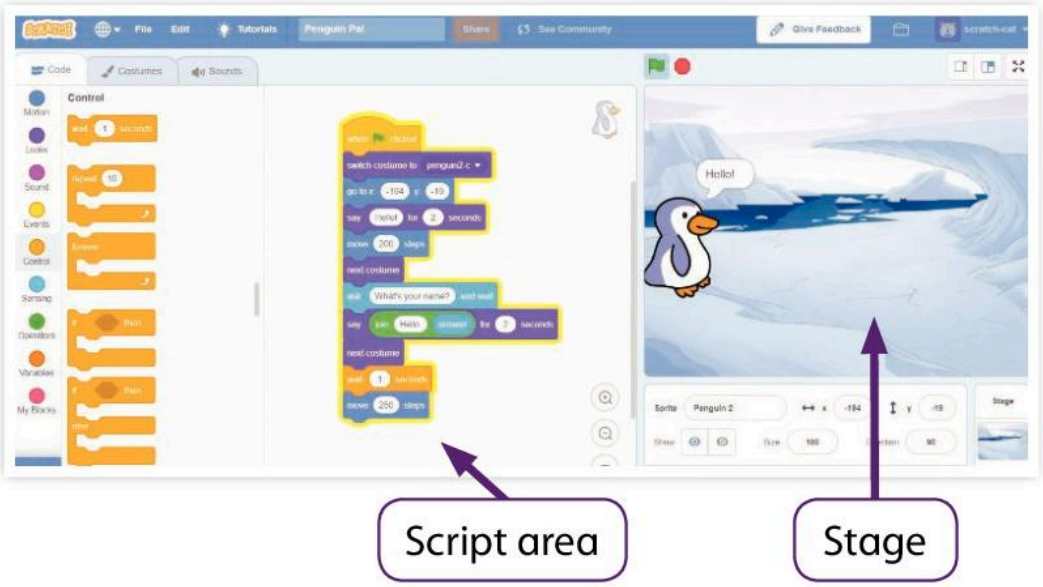
First you must **load** the program. Loading means getting a program from storage. The penguin pal program is stored on your computer.

Open the file menu. Click on 'Load from your computer'. Choose the program called 'Penguin Pal'.



The Scratch screen

You will see a screen like this:



Can you see the **script area**? The script area shows the program.

Can you see the **stage**? The stage is where the sprite moves.

Run the program

Every program script starts with an **'Event' block**. This program starts with a block that shows a green flag. Click on the green flag. This will start the program.

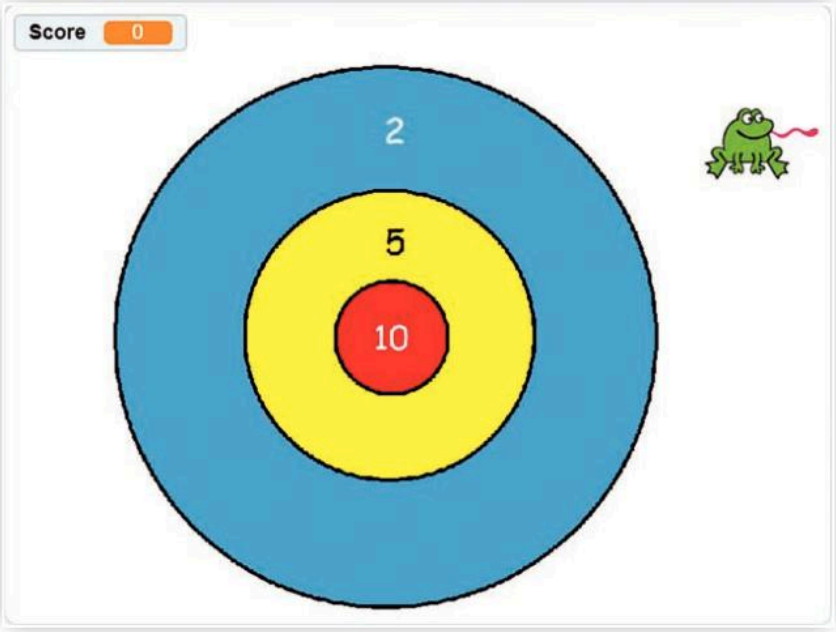


Activity

Load and run the penguin program.



Extra challenge



Use the skills you have learned to load and run the Frog Hop program.



Explore more

Tell your family about the Frog Hop Game. If you get a chance, play the game with your friends at school. Ask everyone what they like best about the game. Ask them what could be better. Note down their answers.

Check what you know

You have learned

- how to plan a program
- what an algorithm is
- what happens when you run a program.



Activity

Here is a plan for the actions in a quiz program. They are not sorted into order.



Tell you if the answer is right

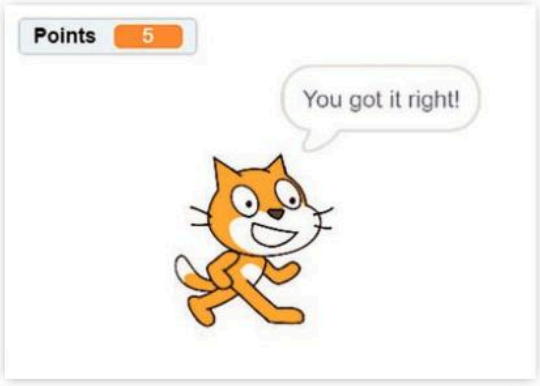
Show how many points you have got

Ask a question

Give you a point for a right answer

Get the answer to the question

Make an algorithm by putting the actions in the right order.



Test

A programmer made a program. Here are the three things she did.

Plan the program
Write the program
Run the program

1. Write the three actions in your book.
2. Tick the action that means carry out the commands.
3. Put a star next to the one that means make an algorithm.
4. Fill in the missing words in these sentences. Use the words from the box.

actions	algorithm	sequence	program

An _____ is a plan to make a _____.

It sets out all the _____ in the right _____.

Self-evaluation

- I answered test questions 1 and 2.
- I answered test questions 1–3.
- I did some of the activity.
- I answered all the test questions.
- I did all of the activity.

Re-read any parts of the unit you feel unsure about. Try the test and activities again – can you do more this time?

4 Programming: The Frog Hop Game

You will learn

- how a program is made of commands
- how to pick commands to make the program you want
- how to make a program that works
- how to make changes and corrections to a program.

In this unit you will make a computer program. You will make a new computer game called the Frog Hop Game. You will use Scratch.



Class activity

Here is a picture of the Scratch screen:



Learning outcomes: Say what a program will do by looking at its commands; Create a simple program and remove errors so it runs

The different parts of the screen are:

- the code blocks: the blocks are stored here ready to use
- the script area: this is where you make a program
- the stage: this is where the sprite moves
- the menu bar: you can choose commands from here.

Point to the different parts of the screen.

Draw the Scratch screen. Write the names of the different parts of the screen.

user **event**
command **forever loop**
random location

Talk about...

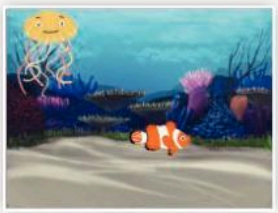
In this unit you will save your work. You must choose a file name. What makes a good file name? Talk to a classmate, then share your ideas with the rest of the class.



Did you know?

Scratch is used by children in more than 150 different countries. Scratch is available in different languages. You can make Scratch programs in more than 40 different languages.









4.1 Choose a sprite

In this lesson

You will learn:

- how to choose a sprite for your program.

Spiral back



Last year you used a simple program. It controlled a sprite. Now you will write a program to control a sprite.

In Unit 3 you played the Frog Hop Game. In this unit you will make a simple version of the game. You can choose any sprite you like. It does not have to be a frog.

You will make a Scratch program. Start on the Scratch website.

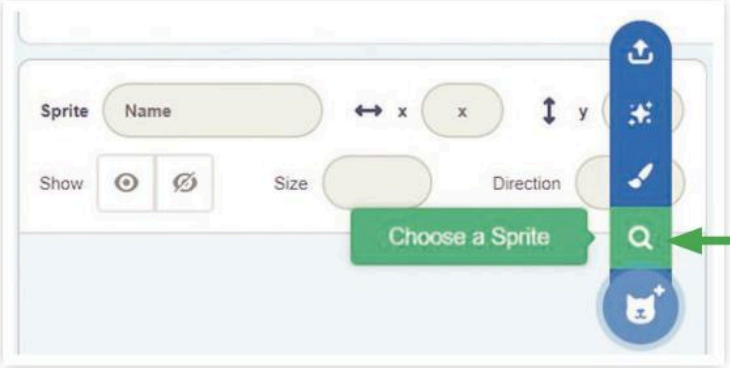
Choose a sprite

Look at the bottom right. This part of the screen shows the sprites.



Sprite 1 is a cat. Delete this sprite by clicking on the cross.

You will choose a new sprite.



Click here to see the other sprites.

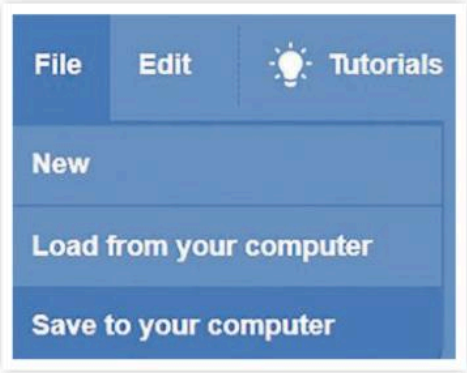
You will see pictures of sprites.



Click on the one you choose.

Save program file

Now you must save your work. You will save it on your computer. Open the File menu and choose 'Save to your computer'. Then type a name for your file.





Activity

Use the skills you learned to:

- pick a sprite
- save your file.



Draw the sprite you chose. Imagine a computer game with your sprite in it. Write down your idea.



Extra challenge

Add a backdrop to your program.

